

Blaque Baux — Financial Methods

A risk-premium harvesting platform: the mathematics, honestly.

Version 1.0 · 2026-07-30 · reference implementation in Julia (execution + strategy) with an archived Python research track.

o. Abstract

Blaque Baux is a systematic, governed trading platform. Its live strategy — the **spine** — does **not** attempt to predict returns. It harvests **risk premia** (diversification, trend, term/asset premia) and manages risk with a stateful daily process, then routes every order through a governed execution layer that enforces hard invariants. This document specifies the financial mathematics as implemented, and states plainly what is validated, what is research, and what did **not** work.

The central empirical result that shaped the platform: on the instruments and horizons tested, **there was no exploitable predictive edge** (information coefficient ≈ 0 ; see §7). The edge that *does* exist and survives out-of-sample is in the **risk structure** — diversification and risk-timing — not in return forecasting. Everything below follows from that finding.

Not investment advice. This is an engineering/quantitative reference. All performance figures are historical backtests or paper-trading results, net of modeled costs; they are not a promise of future results. Trading involves risk of loss.

1. Notation and data

Let there be N assets and discrete time steps $t = 1, \dots, T$ (daily bars unless stated).

- **Prices** $P_{t,i}$ (split/dividend-adjusted, i.e. total-return closes).
- **Simple returns**

$$r_{t,i} = \frac{P_{t,i}}{P_{t-1,i}} - 1.$$

- $R \in \mathbb{R}^{T \times N}$ is the return matrix (rows = time, columns = assets). This $T \times N$ convention is used throughout the code.
- $w_t \in \mathbb{R}^N$ is the target weight vector at time t (fractions of capital; may be negative for shorts; $\sum_i |w_{t,i}|$ is **gross** exposure, $\sum_i w_{t,i}$ is **net**).
- Annualization uses **ppy** periods per year (252 for daily).

Reference universe (US ETFs, chosen for long history and low survivorship bias): **SPY** (US equity), **IEF** (7–10y Treasuries), **TLT** (20y+ Treasuries), **GLD** (gold), **DBC** (broad commodities). The core validation figures below use these five (2007–2026). *Production adds a sixth, **DBA** (agriculture), the one sleeve found to be genuinely uncorrelated (0.44 daily-return corr to DBC vs 0.66–0.84 for tested equity/credit/REIT/silver candidates); on 2016–2026 it lifted the spine's Sharpe 0.94→1.04 at ~flat drawdown — see [CANONICAL_ARCHITECTURE.md](#), "Universe extension." It rests as much on the economic case (weather/supply-driven risk) as on the shorter backtest.*

2. Covariance estimation (EWMA / RiskMetrics)

The strategy is driven by **second moments**, which are more forecastable than first moments. The default estimator is an exponentially-weighted (RiskMetrics-style) covariance with half-life H .

Decay per step and normalized weights over a trailing window of length T :

$$\lambda = 0.5^{1/H}, \quad \tilde{w}_j = \frac{\lambda^{T-j}}{\sum_{k=1}^T \lambda^{T-k}}, \quad j = 1, \dots, T.$$

Weighted mean and covariance:

$$\hat{\mu} = \sum_j \tilde{w}_j R_{j,\cdot}, \quad \hat{\Sigma} = \sum_j \tilde{w}_j (R_{j,\cdot} - \hat{\mu})^\top (R_{j,\cdot} - \hat{\mu}),$$

symmetrized as $\frac{1}{2}(\hat{\Sigma} + \hat{\Sigma}^\top)$. The spine uses $H = 21$ ($\approx$ a pandas `span -60` EWMA), matching the horizon at which the strategy was validated.

Also available (`moments.jl`): unbiased `sample_cov`, Ledoit–Wolf constant-correlation `shrinkage_cov` (analytic optimal intensity), `nearest_psd`, `cov2cor`.

3. The spine — portfolio construction

The spine is a **two-sleeve** book: a long-biased *base* that harvests the diversification premium by risk structure, plus a *trend* sleeve that is the divergent, crisis-hedging component. Each sleeve is independently volatility-targeted, the two are blended, and a regime overlay scales gross exposure.

3.1 Base sleeve — risk-based weights (no return forecast)

The base allocates by **risk**, not expected return. Three options (`riskbased.jl`), default **inverse-vol**:

- **Inverse-volatility** (naive risk parity):

$$w_i^{\text{iv}} = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}, \quad \sigma_i = \sqrt{\hat{\Sigma}_{ii}}.$$

- **Inverse-variance:** $w_i \propto 1/\sigma_i^2$.
- **Equal-risk-contribution (ERC / risk parity):** weights such that every asset contributes equally to portfolio variance. With marginal contribution $\mathbf{RC}_i = w_i(\hat{\Sigma}w)_i$, ERC solves the convex program

$$\min_{w>0} \frac{1}{2} w^\top \hat{\Sigma} w - \sum_i b_i \ln w_i, \quad b_i = \frac{1}{N},$$

whose first-order condition is $w_i(\hat{\Sigma}w)_i = b_i$ for all i . It is solved by cyclical coordinate descent (solver-free, robust inside a backtest loop): for each i , solve the scalar quadratic

$$\hat{\Sigma}_{ii} w_i^2 + \left(\sum_{j \neq i} \hat{\Sigma}_{ij} w_j \right) w_i - b_i = 0, \quad w_i > 0,$$

then renormalize $w \leftarrow w / \sum_i w_i$.

Also implemented: **maximum diversification** (Choueifaty–Coignard: maximize the diversification ratio $\frac{w^\top \sigma}{\sqrt{w^\top \hat{\Sigma} w}}$) and **Hierarchical Risk Parity** (López de Prado: correlation-distance $d_{ij} = \sqrt{(1 - \rho_{ij})/2}$, hierarchical clustering, quasi-diagonalization, recursive bisection by inverse cluster variance — no matrix inversion, stable when N is large relative to sample length).

3.2 Trend sleeve — time-series momentum

The trend sleeve is the only place a *directional* view enters, and only at a horizon where momentum is real (§8.1). For each asset, the 12-month time-series-momentum sign:

$$s_i = \text{sign}\left(\prod_{u=t-k+1}^t (1 + r_{u,i}) - 1\right), \quad k = 252.$$

Inverse-vol scaled and gross-normalized to a long/short book:

$$w_i^{\text{trend}} = \frac{s_i / \sigma_i}{\sum_j |s_j / \sigma_j|}, \quad \sum_i |w_i^{\text{trend}}| = 1.$$

This is the standard managed-futures construction (Moskowitz–Ooi–Pedersen): it goes long assets trending up and **short** assets trending down, sized to equalize each position's risk.

3.3 Per-sleeve volatility targeting (load-bearing)

Each sleeve is scaled to a target volatility σ^* (default 8%), using the sleeve's **own realized P&L volatility**, tracked with a RiskMetrics recursion on the realized sleeve return $p_t = w_{t-1} \cdot r_t$:

$$v_t^2 = \lambda v_{t-1}^2 + (1 - \lambda) p_t^2, \quad \hat{\sigma}_t^{\text{sleeve}} = \sqrt{v_t^2 \cdot \text{ppy}}.$$

Exposure multiplier (capped, de-risk-only when $\text{cap} \leq 1$):

$$e_t = \min\left(\text{cap}, \frac{\sigma^*}{\hat{\sigma}_t^{\text{sleeve}}}\right).$$

Why realized-P&L vol, not ex-ante asset vol? A long/short trend book's P&L can be *smooth while it works* (e.g. steadily short bonds in 2022) even as the underlying asset vols spike. Targeting on ex-ante asset-covariance vol would throttle the hedge exactly when it pays; targeting on the sleeve's own realized P&L keeps it engaged. This distinction is verified — using ex-ante vol collapses the strategy's crisis performance.

3.4 Blend and regime overlay

Blend the vol-targeted sleeves (default $\theta = 0.5$):

$$\tilde{w}_t = \theta e_t^{\text{base}} w_t^{\text{base}} + (1 - \theta) e_t^{\text{trend}} w_t^{\text{trend}}.$$

A discrete **regime brake** scales gross exposure using the drawdown state of an equal-weight market proxy $m_t = \frac{1}{N} \sum_i r_{t,i}$ with index $I_t = \prod_{u \leq t} (1 + m_u)$:

$$\text{dd}_t = \frac{I_t}{\max_{u \leq t} I_u} - 1, \quad g_t = \begin{cases} \phi & \text{if } \text{dd}_t < -0.08 \\ 1 & \text{otherwise} \end{cases} \quad (\phi = 0.5).$$

Final target book:

$$w_t = g_t \cdot \tilde{w}_t.$$

The regime brake was selected empirically over four candidate signals (vol-spike, drawdown, correlation, trend); the drawdown variant improved risk-adjusted return at $\approx$ zero cost, while the others whipsawed or never triggered on a diversified book (§8.2).

3.5 Stateful daily update

In production the spine is stateful: `SpineState` carries each sleeve's RiskMetrics variance ($v^{\text{base}}, v^{\text{trend}}$) and the last-held weights across days. Each trading day, `spine_step!(state, window)`:

1. folds the just-realized bar into each sleeve's vol via §3.3 (using the previously-held weights);
2. recomputes base and trend weights from the trailing window (§3.1–3.2);
3. sizes each sleeve (§3.3), blends (§3.4), applies the regime brake;
4. returns the target book and stores the new sleeve weights.

Positions are then converted to signed share targets $q_i = w_i \cdot C / P_i$ (capital C , price P_i) and reconciled against the broker so the strategy trades the **delta**, not the full target, each day.

4. Risk metrics

Standard definitions (`metrics.jl`), all annualized with $\text{ppy} = 252$.

- **Annualized return** (geometric): $\left(\prod_t(1 + r_t)\right)^{\text{ppy}/T} - 1$.
- **Annualized volatility**: $\text{std}(r)\sqrt{\text{ppy}}$.
- **Sharpe ratio**: $\frac{\bar{r} \text{ppy} - r_f}{\text{std}(r)\sqrt{\text{ppy}}}$.
- **Sortino ratio** (downside deviation about a minimum acceptable return **MAR**):

$$\text{DD} = \sqrt{\text{mean}(\min(r - \text{MAR}, 0)^2)} \sqrt{\text{ppy}}, \quad \text{Sortino} = \frac{\text{AnnRet} - r_f}{\text{DD}}.$$

- **Drawdown series and max drawdown**: with cumulative $C_t = \prod_{u \leq t}(1 + r_u)$,

$$\text{DD}_t = \frac{C_t}{\max_{u \leq t} C_u} - 1, \quad \text{MaxDD} = \min_t \text{DD}_t.$$

- **Calmar ratio**: $\text{AnnRet} / |\text{MaxDD}|$.
- **Value-at-Risk** at confidence α (three methods):
 - *Historical*: $\text{VaR}_\alpha = -Q_{1-\alpha}(r)$.
 - *Gaussian*: $-(\mu + z_{1-\alpha}\sigma)$.
 - *Cornish–Fisher* (skew/excess-kurtosis adjusted):

$$z_{cf} = z + (z^2 - 1)\frac{s}{6} + (z^3 - 3z)\frac{k}{24} - (2z^3 - 5z)\frac{s^2}{36}, \quad \text{VaR}_\alpha = -(\mu + z_{cf}\sigma).$$

- **Expected shortfall (CVaR)**: $\text{ES}_\alpha = -\text{mean}\{r : r \leq Q_{1-\alpha}(r)\}$.

5. Governed execution — risk invariants as guards

Every order passes through one governed path (`ExecutionController`) that enforces hard, testable invariants before the order can reach a venue. These are risk *controls*, not strategy — the strategy proposes, the controller disposes.

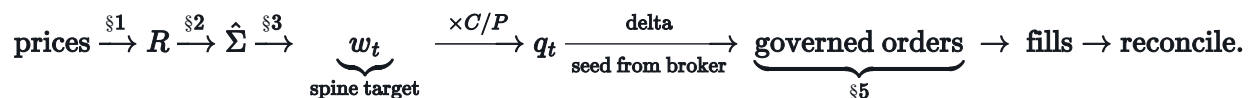
INVARIANT	RULE
Idempotency (REQ-EXEC-002)	Client order id = <code>pool symbol solve_id</code> ; a repeated solve never double-submits (also deduped broker-side).
Per-pool budget (REQ-RISK-003)	Reject if cumulative emitted notional in a pool exceeds its daily budget; reset each trading day.
Daily loss halt (REQ-RISK-004)	Halt the pool if realized daily P&L $\leq -L_{\max}$.
Data staleness (REQ-DATA-003)	Reject emission if the pool's market data is older than a threshold.
Reconciliation (REQ-EXEC-003)	Compare fill-driven expected positions to broker positions; on divergence beyond tolerance, halt.
Kill switch (REQ-GOV-002)	A manual/alert stop halts all normal emission; liquidation orders bypass it (you must always be able to flatten).
Fill lineage (REQ-AUDIT-001)	Every recorded fill carries <code>signal_id / regime / solve_id / order_id</code> .

5.1 Layer-3 pre-trade safety gate (live money)

Before any order in live mode, a pre-flight aggregate must pass (`SafetyGate.preflight`), else the book is halted and an alert fires. Guards include:

- **Drawdown halt:** with a persisted equity high-water mark **HWM**, halt if $\frac{E}{\text{HWM}} - 1 < -d_{\max}$ (default 15%).
- **Daily-loss halt:** halt if $E - E_{\text{prev}} < -L_{\max}$.
- **Gross-leverage cap:** reject if $\frac{\sum_i |q_i| P_i}{E} > \Lambda_{\max}$ (default 2×).
- **Per-name cap:** reject if any $|q_i| P_i / E$ exceeds a bound (default 0.85 — a *runaway-bug* catcher; genuine concentration risk is handled by the risk-based construction and the gross/drawdown caps, since inverse-vol legitimately places large weight in low-vol bonds).
- **Account & data checks:** account active/not blocked, sufficient buying power, data fresh.

6. Composition — the daily pipeline



One rebalance per trading day (§8.3 explains why not intraday), executed pre-open so orders fill at the open.

7. Validation methodology — measuring edge honestly

The platform's design was gated on **out-of-sample edge tests**, not in-sample fit.

7.1 Information Coefficient

For a per-bar cross-sectional score $x_{t,i}$ and forward return $r_{t+1,i}$, the **Information Coefficient** is the Spearman rank correlation across assets each window; the **IC information ratio** is its t -statistic across N_w windows:

$$IC_t = \rho_{\text{Spearman}}(x_{t,\cdot}, r_{t+1,\cdot}), \quad IC\text{-}IR = \frac{\overline{IC}}{\text{std}(IC)} \sqrt{N_w}.$$

A signal is treated as real only if roughly $|\overline{IC}| > 0.02$ and $|IC\text{-}IR| > 2$.

7.2 Out-of-sample discipline

- **Train/test split** and walk-forward evaluation; parameters fixed *a priori* (round numbers), never tuned on the test window.
- **Control for confounds:** e.g. when testing a regime-conditional blend, compare not only to the static baseline but to *static-at-the-adaptive-rule's-own-mean-weight*, to separate genuine timing value from "just ran a different constant."
- **Purged K-fold / combinatorial-purged CV** (López de Prado) is available for series with overlapping labels (`module_11_cv`).
- **Multiple-testing awareness:** a rule is granted one or two well-motivated shots; fishing variants until one passes is treated as p-hacking and rejected.

7.3 The decisive result

On 60-day 15-minute data ($\approx 1,547$ bars $\times$ 60 S&P names), two candidate alpha models were tested:

MODEL	DIRECTIONAL ACCURACY	P VS 50%	IC
3-factor cross-sectional	50.9%	0.48	-0.004
Bayesian (Kalman/NIG/James–Stein/ADVI)	50.7%	0.59	+0.001

Both are **coin flips**. There was no exploitable predictive edge at this horizon/universe. This is *why* the platform harvests risk premia rather than forecasts returns.

8. Empirical findings that shaped the design

All figures are historical backtests, net of ~ 2 bps/side costs, on the reference universe.

8.1 Momentum vs. reversal by horizon

IC of a trailing k -day return against the next-day return, pooled across assets, 2006–2026:

K	1D	2D	3D	5D	10D	21D	63D	126D	252D
IC	−.038	−.041	−.035	−.031	−.021	−.010	−.000	+.005	+.002

Short horizons (1–63d) exhibit **reversal** (a sharp drop tends to bounce); **momentum** appears only at 126–252d. This is the quantitative reason the trend sleeve uses a **12-month** lookback and why "shorting a 2-day selloff" fights the strongest short-horizon force in the data.

8.2 The validated spine

Base ⊕ trend ⊕ vol-target ⊕ regime brake, 2007–2026 (incl. GFC, COVID, 2022):

CONFIGURATION	SHARPE	SORTINO	MAX DD	2022
Diversified base only	0.85	1.24	−18%	−15%
Spine (base + trend), regime : none	0.94	1.28	−10.6%	−3.2%
Spine, regime : dd (production)	0.97	1.30	−10.6%	−1.4%

The trend sleeve made **+4.4% in 2022** (short bonds / long energy) while a long-only book bled — that is the crisis hedge working as designed.

8.3 Cadence — daily, not intraday

On 17,659 real 15-minute bars (2022–2026), applying the daily book intraday: holding through the day gives Sharpe ≈ 1.24 . An intraday vol-brake **never triggered** (a 5-asset diversified book barely moves intraday — stocks/bonds/gold/commodities offset within the bar; diversification is the intraday shock absorber). An intraday drawdown-brake **whipsawed** (−1% CAGR for −0.7% max-DD). Full 15-minute re-optimization is pure cost drag — risk premia do not change in 15 minutes.
Conclusion: once-daily rebalance.

8.4 Leverage — the return/drawdown trade-off

Leverage L on the strategy return r_t , financing the borrowed portion at annual rate f :

$$r_t^L = L r_t - (L - 1) \frac{f}{\text{ppy}}.$$

2007–2026, financing at 3% (the friendliest case) vs. 6.5% (today's retail margin):

L	CAGR @3%	CAGR @6.5%	VOL	SHARPE	MAX DD
1.0	+5.9%	+5.9%	6.1%	0.97	−10.6%
1.5	+7.2%	+5.4%	9.2%	0.81	−18.1%
2.0	+8.5%	+4.7%	12.2%	0.73	−25.1%
3.0	+10.6%	+3.2%	18.3%	0.64	−39.1%

Two facts: (i) **Sharpe only falls with leverage** (financing is a pure drag on risk-adjusted return); (ii) at today's ~6.5% margin, leverage is **net-negative** — the strategy's ~6% return is below the cost of borrowing, so leverage subtracts return while amplifying drawdown roughly $\propto L$. Double-digit CAGR requires ~3×, which comes with a **−39% drawdown**. There is no double-digit-return-at-low-risk configuration.

9. The portfolio-optimization library — optimizers, Monte-Carlo robustification, and a service

The spine (§3) is one strategy built on a **general-purpose optimization library**.

`module_13_portfolio (PortfolioOpt)` is usable on its own: every optimizer consumes the moments/covariance of §1–2 (or a raw return matrix) and returns budget-constrained weights. Convex programs are solved with **JuMP** and the **Clarabel** conic solver; a long-only toggle and scalar/vector box bounds are shared across the family.

Caveat, stated up front. Any optimizer that consumes expected returns μ (§9.1, §9.2) inherits the Markowitz *error-maximization* problem — it over-fits noise in the sample mean. That is *why* the flagship spine avoids μ entirely (§3), and why this library ships the antidotes next to the optimizers: shrinkage (§2), equilibrium anchoring (§9.2), and Monte-Carlo resampling (§9.5).

9.1 Markowitz family (`meanvariance.jl`)

Quadratic programs over $\sum_i w_i = 1$ with optional long-only and box constraints:

- **Minimum variance:** $\min_w w^\top \hat{\Sigma} w$ — the one member that uses no μ , hence the robust one.
- **Maximum Sharpe (tangency):** $\max_w \frac{\mu^\top w - r_f}{\sqrt{w^\top \hat{\Sigma} w}}$, via the standard convex reformulation.
- **Mean-variance:** $\max_w \mu^\top w - \frac{\gamma}{2} w^\top \hat{\Sigma} w$ (risk-aversion γ) or $\min_w w^\top \hat{\Sigma} w$ s.t. $\mu^\top w \geq r^*$ (target-return).
- **Efficient frontier:** the target-return sweep across n points.

9.2 Black–Litterman (`blacklitterman.jl`)

Anchor on the market and blend in views, rather than trusting raw sample means. Reverse-optimize the equilibrium prior from market-cap weights,

$$\Pi = \delta \hat{\Sigma} w_{\text{mkt}},$$

then form the posterior from K views (P, Q) with view-uncertainty Ω and prior scaling τ :

$$\mu_{\text{BL}} = \left[(\tau \hat{\Sigma})^{-1} + P^{\top} \Omega^{-1} P \right]^{-1} \left[(\tau \hat{\Sigma})^{-1} \Pi + P^{\top} \Omega^{-1} Q \right], \quad \Sigma_{\text{BL}} = \hat{\Sigma} + \left[(\tau \hat{\Sigma})^{-1} + P^{\top} \Omega^{-1} P \right]^{-1}.$$

`make_view` builds a row of P with its Q entry ("these assets will return q "); the posterior $(\mu_{\text{BL}}, \Sigma_{\text{BL}})$ then feeds any §9.1 optimizer. Omitting Ω defaults it to $\text{diag}(P \tau \hat{\Sigma} P^{\top})$.

9.3 Tail-risk optimizers (`tailrisk.jl`)

These shape the *realized* loss distribution from the scenario matrix R (not a Gaussian proxy).

Minimum CVaR uses the Rockafellar–Uryasev linear program: at level α ,

$$\min_{w, \zeta, u} \zeta + \frac{1}{(1 - \alpha)T} \sum_{t=1}^T u_t \quad \text{s.t.} \quad u_t \geq -R_{t, \cdot} w - \zeta, \quad u_t \geq 0, \quad \sum_i w_i = 1,$$

optionally with a return floor $\mu^{\top} w \geq r^*$. It remains an **LP** regardless of scenario count. **Minimum CDaR** is the drawdown analogue (conditional drawdown-at-risk).

9.4 Cost-aware optimization (`costaware.jl`)

Decide what to *trade to* given current holdings w_0 , so cost shapes the solution instead of being measured afterward:

$$\text{turnover} = \sum_i |w_i - w_{0,i}|, \quad \text{cost}(w_0 \rightarrow w) = c_{\text{lin}} \sum_i |\Delta_i| + c_{\text{imp}} \sum_i \Delta_i^2, \quad \Delta_i = w_i - w_{0,i},$$

and `mean_variance_tc` maximizes $\mu^{\top} w - \frac{\gamma}{2} w^{\top} \hat{\Sigma} w - \text{cost}(w_0 \rightarrow w)$ — a linear (spread/commission) plus convex (market-impact) penalty *inside* the objective.

9.5 Monte-Carlo robustification (`robust.jl`)

This is the platform's Monte Carlo — used to **defend against estimation error**, not to forecast price. A data-generating process is a pluggable sampler `(R, T_sim) -> simulated returns`:

- `gaussian_dgp` — draw from $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$;
- `iid_bootstrap` — resample whole rows with replacement (nonparametric; preserves the cross-sectional dependence);
- `block_bootstrap(block)` — circular blocks (preserves short-horizon autocorrelation);
- `stationary_bootstrap(meanblock)` — Politis–Romano geometric block lengths;
- `student_t_dgp(v)` — fat-tailed multivariate- t scaled to $\hat{\Sigma}$.

Two consumers:

- **Feasible-set Monte Carlo** (`random_portfolios`): sample many admissible weight vectors to trace the achievable risk/return cloud — the dashboard's scatter, and a sanity check on any optimizer's output.
- **Resampled (Michaud) efficient frontier** (`resampled_frontier`): for each of n_{resample} draws, simulate T_{sim} returns from the sampler, compute that draw's frontier, and **average the weights** across draws. The averaged frontier is markedly more stable out-of-sample than the single-shot Markowitz frontier — the direct antidote to §9.1's error-maximization.

9.6 The optimizer as a service (`scripts/`)

The whole stack is exposed over HTTP for a dashboard or an external caller — `portfolio_server.jl` (JSON REST, default `127.0.0.1:8766`) with a Python **Dash** frontend (`dashboard.py`):

ENDPOINT	RETURNS
POST <code>/optimize</code>	one strategy's weights (min-var, max-Sharpe, risk-parity, min-CVaR, ...)
POST <code>/frontier</code>	efficient frontier (n points)
POST <code>/resampled_frontier</code>	Michaud resampled frontier (§9.5)
POST <code>/backtest</code>	walk-forward backtest of a chosen strategy
POST <code>/metrics</code>	performance metrics (§4) for a return series
POST <code>/random_portfolios</code>	feasible-set Monte-Carlo cloud (§9.5)
GET <code>/health</code>	liveness

```
julia --project=. scripts/portfolio_server.jl      # backend on :8766
python scripts/dashboard.py                      # Dash UI on :8050
```

This makes the repository usable as a **general portfolio-optimization library and service**, with the spine as the flagship strategy built on top of it. (A production data layer — `module_1_data/data_feeds_production.jl` — can feed it live inputs including a **Deribit BTC volatility-index** signal; crypto enters as a risk input, not as a traded sleeve.)

10. The research track (Python) — archived, and why

The platform began as an alpha-prediction system. That code is preserved (archived) because the mathematics is sound, but it is **not on the live path** — edge validation (§7) found no predictive signal at the horizons tested. Documented honestly:

- **Bayesian alpha engine** (`signal_engine.py`): per-ticker return estimation via Kalman filtering → Normal-Inverse-Gamma conjugate updating → James–Stein shrinkage → ADVI, combined by inverse-variance weighting. Mathematically correct; **no measured predictive edge** (§7.3).
- **Monte-Carlo risk engine** (`monte_carlo_engine.py`): geometric-Brownian-motion simulation of a portfolio given *assumed* drift and volatility, producing VaR/CVaR/expected-shortfall/probability-of-profit. It is a **risk calculator, not an optimizer** — it sizes a position's tail; it does not create return. Correct tool for sizing a premium by its tail; wrong tool to find one.
- **Six-Sigma Oracle** (`risk_engine.py`): samples weight vectors and minimizes a simulated loss rate subject to a return target — i.e. Sharpe maximization on **historical means**. This is the Markowitz *error-maximization* trap: optimizing on noisy sample means fails out-of-sample. Its correct use is **survival-sizing** ("at exposure **X**, what is the six-sigma loss?" → cap gross so the tail stays within budget), not return generation.

The lesson, stated once: sophisticated machinery (long/short, QP optimization, Monte-Carlo, Bayesian filtering) *packages and sizes* a signal; it does not *create* one. The signal is the constraint, and at the tested horizons it was not there.

11. Code map

CONCEPT (§)	FILE · KEY SYMBOLS
Returns, moments, EWMA cov (§1–2)	<code>src/module_13_portfolio/moments.jl</code> · <code>simple_returns</code> , <code>ewma_cov</code> , <code>shrinkage_cov</code> , <code>nearest_psd</code>
Base weights (§3.1)	<code>src/module_13_portfolio/riskbased.jl</code> · <code>inverse_variance</code> , <code>risk_parity</code> , <code>max_diversification</code> , <code>hrp_weights</code>
Spine: trend, vol-target, blend, regime, stateful (§3.2–3.5)	<code>src/module_13_portfolio/spine.jl</code> · <code>tsmom_signal</code> , <code>tsmom_weights</code> , <code>voltarget_exposure</code> , <code>regime_multiplier</code> , <code>SpineState</code> , <code>spine_step!</code> , <code>spine_targets</code>
Metrics, VaR/CVaR (§4)	<code>src/module_13_portfolio/metrics.jl</code> · <code>sharpe</code> , <code>sortino</code> , <code>max_drawdown</code> , <code>calmar</code> , <code>value_at_risk</code> , <code>expected_shortfall</code>
Optimizer library — Markowitz / BL / tail-risk / cost-aware (§9.1–9.4)	<code>meanvariance.jl</code> · <code>min_variance</code> , <code>max_sharpe</code> , <code>mean_variance</code> , <code>efficient_frontier</code> ; <code>blacklitterman.jl</code> · <code>black_litterman</code> , <code>implied_equilibrium_returns</code> , <code>make_view</code> ; <code>tailrisk.jl</code> · <code>min_cvar</code> , <code>min_cdar</code> ; <code>costaware.jl</code> · <code>mean_variance_tc</code> , <code>transaction_cost</code> , <code>turnover</code>
Monte-Carlo robustification (§9.5)	<code>src/module_13_portfolio/robust.jl</code> · <code>gaussian_dgp</code> , <code>iid_bootstrap</code> , <code>block_bootstrap</code> , <code>stationary_bootstrap</code> , <code>student_t_dgp</code> , <code>random_portfolios</code> , <code>resampled_frontier</code>
Optimizer REST service + dashboard (§9.6)	<code>scripts/portfolio_server.jl</code> , <code>scripts/dashboard.py</code>
Production data feeds (Cboe / FRED / Deribit crypto-vol)	<code>src/module_1_data/data_feeds_production.jl</code>
Purged / combinatorial CV (§7.2)	<code>src/module_11_cv/</code>
Governed execution invariants (§5)	<code>src/module_7_execution/</code> · <code>execution_controller.jl</code> , <code>venue_interface.jl</code> , <code>venues/</code>
Layer-3 safety gate (§5.1)	<code>src/module_8_governance/safety_gate.jl</code> · <code>SafetyLimits</code> , <code>preflight</code> , <code>drawdown</code>
Data adapters (Alpaca / IBKR / CSV)	<code>src/module_1_data/</code> · <code>equity_panel.jl</code> , <code>alpaca_panel.jl</code> , <code>ibkr_panel.jl</code>
Live driver + safety	<code>scripts/spine_live.jl</code> , <code>scripts/run_spine_daily.sh</code>
Leverage / cadence / validation analyses (§7–8)	<code>scripts/leverage_decision_data.jl</code> , <code>docs/CANONICAL_ARCHITECTURE.md</code> , <code>docs/leverage_decision.html</code>

CONCEPT (§)	FILE · KEY SYMBOLS
Research track (archived)	Archive/blaque_baux/... · signal_engine.py , optimizer/risk_engine.py ; Archive/crypto-quant-mvp-v2/monte_carlo_engine.py

12. References

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13. Disclaimer

This document and the accompanying code are provided for **educational and research purposes**. They do not constitute investment advice, a solicitation, or an offer. Performance figures are historical simulations or paper-trading results, net of modeled transaction costs and (where noted) financing; they are **not indicative of future results**. Systematic trading carries substantial risk, including loss of principal. Anyone deploying real capital does so at their own risk and should conduct independent validation.